PROBABILITY · STATISTICS · DATA SCIENCE

Probability and Mathematical Statistics

Problem Book: From Fundamentals to Admissions Challenges

144 graded problems · hints and answers · 24 selected solutions

This companion volume is organized as a deliberate progression: direct uses of definitions, standard methods, multi-step synthesis, and concise challenge problems in the style of selective data-science admissions examinations.

How to train with this book

The problems are synchronized with the 24 chapters of the theory volume. Each chapter contains six problems.

Level	Purpose	Suggested time
Warm-up	Translate definitions into a calculation	5–10 min
Core	Select and execute one standard method	10–20 min
Advanced	Combine two or more ideas	20–35 min
Challenge	Invent a reduction or proof	35–50 min

For every problem, first write the random experiment and the target quantity. Use a hint only after a documented attempt. Read the answer after completing a proof, not merely after obtaining a plausible number. The selected solutions demonstrate the expected standard of exposition; they are not templates to memorize word for word.

You have mastered a chapter when you can solve both Core problems without notes, at least one Advanced problem within 30 minutes, and explain the key idea of the Challenge problem the next day from memory.

Contents

Part I

Probability Foundations

CHAPTER 1

Counting Before Probability

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 1.1 · Warm-up — Constrained strings

How many length-12 binary strings contain exactly four ones? How many additionally have no adjacent ones?

Problem 1.2 · Core — Committee with restrictions

A committee of five is selected from eight mathematicians and six programmers. How many committees contain at least two people from each group?

Problem 1.3 · Core — Integer solutions

Count the positive integer solutions of $x_1 + x_2 + x_3 + x_4 = 20$ subject to $x_1 \geq 3$.

Problem 1.4 · Advanced — A double count

Prove $\sum_{k=0}^{n} k\binom{n}{k} = n2^{n-1}$ without differentiating the binomial theorem.

Problem 1.5 · Advanced — Onto functions

How many functions from an n-element set onto a three-element set are there?

Problem 1.6 · Challenge — Derangements recurrence

Let D_n be the number of derangements of n objects. Prove $D_n = (n-1)(D_{n-1} + D_{n-2})$ and use it to compute D_6.

CHAPTER 2

Probability Spaces and Events

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 2.1 · Warm-up — Cards

Five cards are drawn uniformly from a standard 52-card deck. Find the probability of exactly two aces.

Problem 2.2 · Core — First collision

Four people have independent uniform birthdays among 365 days. Find the probability that all birthdays are distinct.

Problem 2.3 · Core — Random point in a disk

A point is uniform in the unit disk. Find $\mathbb{P}(R \le r)$ and the density of its distance R from the center.

Problem 2.4 · Advanced — Continuity from below

Let $A_n = \{X \le n\}$. Prove $\mathbb{P}(A_n) \to 1$ for every finite real-valued random variable X.

Problem 2.5 · Advanced — A nonuniform mechanism

An integer is generated by choosing its number of decimal digits uniformly from $\{1, 2\}$, then choosing a valid integer of that length uniformly. Compare the probabilities of 7 and 70.

Problem 2.6 · Challenge — Broken stick

Two points are chosen independently and uniformly on a unit segment. The three pieces form a triangle exactly when no piece exceeds $1/2$. Find the probability.

CHAPTER 3

Conditional Probability and Bayes' Rule

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 3.1 · Warm-up — Two draws

An urn contains three white and two black balls. Two are drawn without replacement. Find the probability both are white.

Problem 3.2 · Core — A selected coin

One coin is fair and one has heads on both sides. A coin is selected uniformly and shows heads. Find the posterior probability that it is double-headed.

Problem 3.3 · Core — Medical screening

A condition has prevalence 0.5%. A test has sensitivity 98% and specificity 95%. Find the positive predictive value.

Problem 3.4 · Advanced — Repeated evidence

A hidden coin is fair with prior probability $2/3$ and has head probability $3/4$ otherwise. Given three heads in three tosses, find the posterior probability that it is the biased coin.

Problem 3.5 · Advanced — Even waiting time

A fair die is rolled until the first six. Find the probability that the number of rolls is even.

Problem 3.6 · Challenge — Information protocol

One of prisoners A, B, C will be released uniformly. A guard truthfully names one of B, C who will not be released and says B. Show that the posterior probability for A cannot be determined unless the guard's rule is specified; compute it if the guard chooses uniformly when both names are available.

CHAPTER 4

Independence

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 4.1 · Warm-up — Complements

If events A and B are independent, prove that A^c and B are independent.

Problem 4.2 · Core — At least one success

How many independent trials with success probability 0.2 are needed so that the probability of at least one success is at least 0.99?

Problem 4.3 · Core — Pairwise independence

Let X, Y be independent fair bits and $Z = X \oplus Y$. Verify pairwise independence of X, Y, Z and show they are not mutually independent.

Problem 4.4 · Advanced — Conditional dependence

Two fair coin tosses X, Y are independent. Show that they become dependent conditional on $X + Y = 1$.

Problem 4.5 · Advanced — A binomial mode

For $S \sim \mathrm{Bin}(n, p)$, derive the ratio $\mathbb{P}(S = k+1)/\mathbb{P}(S = k)$ and determine the modal value(s).

Problem 4.6 · Challenge — Possible triple intersections

Events A, B, C are pairwise independent and each has probability $1/2$. Determine all possible values of $t = \mathbb{P}(A \cap B \cap C)$.

Part II

Random Variables and Their Calculus

CHAPTER 5

Random Variables and Distribution Functions

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 5.1 · Warm-up — Normalize a density

Let $f(x) = c(1 - x^2)$ for $-1 \le x \le 1$ and zero otherwise. Find c and the CDF.

Problem 5.2 · Core — A mixed law

A variable equals zero with probability $1/3$ and otherwise is uniform on $[0, 1]$. Write its CDF and identify its jump.

Problem 5.3 · Core — Absolute value

Express the CDF of $Y = |X|$ in terms of a continuous CDF F_X. Simplify if X is symmetric about zero.

Problem 5.4 · Advanced — Reciprocal transformation

If $X \sim \mathrm{Unif}[1, 2]$ and $Y = 1/X$, find the density of Y.

Problem 5.5 · Advanced — Quantile transform

Let $U \sim \mathrm{Unif}[0, 1]$. Prove that $X = -\lambda^{-1} \log(1 - U)$ is $\mathrm{Exp}(\lambda)$.

Problem 5.6 · Challenge — Same distribution, different dependence

Construct random variables X, Y with the same $\mathrm{Unif}[0, 1]$ marginal distribution in two models: one where $X = Y$ almost surely and one where X, Y are independent. Compare the distribution of $X - Y$.

CHAPTER 6

Expectation and Variance

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 6.1 · Warm-up — Fixed points

Find the expected number of fixed points in a uniform random permutation of n elements.

Problem 6.2 · Core — Inversions

Find the expected number of inversions in a uniform random permutation of n elements.

Problem 6.3 · Core — Tail sum

A nonnegative integer-valued X satisfies $\mathbb{P}(X \geq k) = 2^{-(k-1)}$ for $k \geq 1$. Find $\mathbb{E}X$.

Problem 6.4 · Advanced — Variance of a sum

Variables X, Y have variances $4, 9$ and correlation $-1/3$. Find $\mathrm{Var}(2X - Y)$.

Problem 6.5 · Advanced — A probabilistic existence proof

Prove that every finite graph has a cut containing at least half its edges.

Problem 6.6 · Challenge — Variance of fixed points

Let F be the number of fixed points in a uniform random permutation of $n \geq 2$ elements. Compute $\mathrm{Var}(F)$.

CHAPTER 7

Canonical Discrete Distributions

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 7.1 · Warm-up — Geometric mean

A die is rolled until a six appears. Find the mean and variance of the number of rolls.

Problem 7.2 · Core — Hypergeometric expectation

A sample of 10 items is drawn without replacement from 100 items, 20 defective. Find the expected number defective and its variance.

Problem 7.3 · Core — Poisson recurrence

For $X \sim \mathrm{Pois}(4)$, determine the modal value(s) without evaluating exponentials.

Problem 7.4 · Advanced — Negative binomial

A coin with head probability p is tossed until the third head. Find the probability of stopping at toss k and the expected stopping time.

Problem 7.5 · Advanced — Poisson approximation

Among 1000 components, each fails independently with probability 0.002. Approximate the probability of at least three failures.

Problem 7.6 · Challenge — Poisson splitting

Let $N \sim \mathrm{Pois}(\lambda)$. Independently label each of the N items type A with probability p and type B otherwise. Prove the two type counts are independent Poisson variables.

CHAPTER 8

Canonical Continuous Distributions

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 8.1 · Warm-up — Uniform moments

For $X \sim \mathrm{Unif}[a, b]$, derive $\mathbb{E}X$ and $\mathrm{Var}(X)$ by integration.

Problem 8.2 · Core — Competing clocks

Independent clocks ring after $X \sim \mathrm{Exp}(2)$ and $Y \sim \mathrm{Exp}(3)$. Find the law of $\min(X, Y)$ and $\mathbb{P}(X < Y)$.

Problem 8.3 · Core — Gamma sum

Let X_1, X_2, X_3 be independent $\mathrm{Exp}(\lambda)$. Write the density, mean, and variance of their sum.

Problem 8.4 · Advanced — Normal standardization

If $X \sim \mathcal{N}(10, 4)$, express $\mathbb{P}(8 \leq X \leq 13)$ using the standard normal CDF.

Problem 8.5 · Advanced — Maximum entropy check

Among densities with fixed mean μ and variance σ^2, the normal maximizes differential entropy. Explain why this supports, but does not prove, using a normal model when only those moments are known.

Problem 8.6 · Challenge — Ratio of normals

Let X, Y be independent standard normal variables. Show that X/Y has the standard Cauchy density.

CHAPTER 9

Joint Distributions and Random Vectors

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 9.1 · Warm-up — A triangular support

A joint density is constant on $0 < x < y < 1$. Find the constant and both marginal densities.

Problem 9.2 · Core — Covariance from a table

The pair (X, Y) is uniform on $\{(0,0),(0,1),(1,0)\}$. Find $\mathrm{Cov}(X, Y)$ and decide whether X, Y are independent.

Problem 9.3 · Core — Linear forms

A random vector has covariance matrix $\Sigma = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}$. Find $\mathrm{Var}(2X - 3Y)$.

Problem 9.4 · Advanced — Uncorrelated dependence

Let $X \sim \mathrm{Unif}[-1, 1]$ and $Y = X^2$. Prove $\mathrm{Cov}(X, Y) = 0$ and explain why they are dependent.

Problem 9.5 · Advanced — Optimal combination

Unbiased estimators T_1, T_2 have variances $1, 4$ and covariance 0. Find the minimum-variance unbiased estimator of the form $aT_1 + (1-a)T_2$.

Problem 9.6 · Challenge — A covariance matrix criterion

For which real r can $\begin{pmatrix} 1 & r & r \\ r & 1 & r \\ r & r & 1 \end{pmatrix}$ be a correlation matrix?

CHAPTER 10

Transformations and Order Statistics

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 10.1 · Warm-up — Maximum of uniforms

For iid $X_1, \dots, X_n \sim \mathrm{Unif}[0, 1]$, find the CDF, density, and mean of $M = \max_i X_i$.

Problem 10.2 · Core — A convolution

Find the density of the sum of independent $\mathrm{Unif}[0, 1]$ variables.

Problem 10.3 · Core — Exponential minimum

If X_i are independent $\mathrm{Exp}(\lambda_i)$, find the law of their minimum and the probability that X_j is the minimum.

Problem 10.4 · Advanced — Median density

For five iid observations with continuous CDF F and density f, find the density of the sample median.

Problem 10.5 · Advanced — Random sum variance

Let N be independent of iid X_i, with $\mathbb{E}N = 4$, $\mathrm{Var}(N) = 3$, $\mathbb{E}X_i = 2$, $\mathrm{Var}(X_i) = 5$. Find the mean and variance of S_N.

Problem 10.6 · Challenge — Independence after a change of variables

Independent $X, Y \sim \mathrm{Exp}(1)$ are transformed to $U = X + Y$ and $V = X/(X + Y)$. Prove U, V are independent and identify their laws.

CHAPTER 11

Conditional Expectation

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 11.1 · Warm-up — A random number of tosses

A fair die is rolled and then a fair coin is tossed that many times. Find the expected number of heads.

Problem 11.2 · Core — Total variance

A population is selected with equal probability. In group 1, X has mean 0 and variance 1; in group 2, mean 3 and variance 4. Find $\mathbb{E}X$ and $\mathrm{Var}(X)$.

Problem 11.3 · Core — Poisson mixture

Let Λ have mean 2 and variance 3, and $X \mid \Lambda \sim \mathrm{Pois}(\Lambda)$. Find $\mathbb{E}X$ and $\mathrm{Var}(X)$.

Problem 11.4 · Advanced — Projection identity

Prove that for $m(Y) = \mathbb{E}[X \mid Y]$ and every square-integrable $g(Y)$, $\mathbb{E}[(X - m(Y))g(Y)] = 0$.

Problem 11.5 · Advanced — Records

In a uniform random permutation of n distinct numbers, a position is a record if its value exceeds all preceding values. Find the expected number of records.

Problem 11.6 · Challenge — Waiting for HTH

A fair coin is tossed until the pattern HTH first appears. Find the expected number of tosses.

Part III

Bounds, Asymptotics, and Stochastic Models

CHAPTER 12

Probability Inequalities

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 12.1 · Warm-up — Markov

A nonnegative variable has mean 12. Give the best Markov bound for $\mathbb{P}(X \geq 60)$.

Problem 12.2 · Core — Chebyshev sample size

Independent observations have variance 9. How large must n be for Chebyshev to guarantee $\mathbb{P}(|\overline{X} - \mu| < 0.5) \geq 0.99$?

Problem 12.3 · Core — Jensen

For positive X, prove $\mathbb{E}X\,\mathbb{E}(1/X) \geq 1$ and characterize equality.

Problem 12.4 · Advanced — Birthday union bound

Use the union bound to upper-bound the probability of a birthday collision among n people under the uniform 365-day model.

Problem 12.5 · Advanced — Chernoff template

For $S \sim \mathrm{Bin}(n, p)$ and $a > np$, derive $\mathbb{P}(S \geq a) \leq \inf_{t>0} e^{-ta}(1 - p + pe^t)^n$.

Problem 12.6 · Challenge — A large independent set

Prove every graph with n vertices and average degree d contains an independent set of size at least $n/(d + 1)$.

CHAPTER 13

Generating and Characteristic Functions

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 13.1 · Warm-up — Binomial PGF

Derive the probability-generating function of $X \sim \mathrm{Bin}(n, p)$ and recover its mean.

Problem 13.2 · Core — Poisson addition

Use PGFs to prove that independent $\mathrm{Pois}(\lambda)$ and $\mathrm{Pois}(\mu)$ variables sum to $\mathrm{Pois}(\lambda + \mu)$.

Problem 13.3 · Core — Compound Poisson

Let $N \sim \mathrm{Pois}(\lambda)$ and, independently, X_i have PGF G. Find the PGF of $S = \sum_{i=1}^{N} X_i$.

Problem 13.4 · Advanced — Moment extraction

A nonnegative integer-valued variable has PGF $G(s) = (1-q)/(1-qs)$, $0 < q < 1$. Identify the law and compute its mean and variance.

Problem 13.5 · Advanced — Poisson limit

Show from PGFs that $\mathrm{Bin}(n, \lambda/n)$ converges in distribution to $\mathrm{Pois}(\lambda)$.

Problem 13.6 · Challenge — Branching extinction

In a Galton–Watson process each individual has offspring PGF G. Prove the extinction probability q is the smallest fixed point of G in $[0, 1]$.

CHAPTER 14

Convergence and Limit Theorems

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 14.1 · Warm-up — Weak law bound

Iid variables have variance 16. Bound $\mathbb{P}(|\overline{X}_n - \mu| \geq 1)$ and find n making the bound at most 0.01.

Problem 14.2 · Core — Convergence without mean convergence

Let $X_n = n$ with probability $1/n$ and zero otherwise. Determine convergence in probability and the behavior of $\mathbb{E}X_n$.

Problem 14.3 · Core — Normal approximation

For $S \sim \mathrm{Bin}(100, 1/2)$, approximate $\mathbb{P}(45 \leq S \leq 55)$ using a continuity correction.

Problem 14.4 · Advanced — Continuous mapping

If $T_n \to \theta > 0$ in probability, prove $1/T_n \to 1/\theta$ in probability.

Problem 14.5 · Advanced — CLT scaling

Iid variables have mean 3 and variance 4. Find the limiting distribution of $\sqrt{n}(\overline{X}_n - 3)$.

Problem 14.6 · Challenge — Distribution but not probability

Give a sequence X_n that converges in distribution but not in probability to a random variable with the same nondegenerate law.

CHAPTER 15

Random Walks and Markov Chains

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 15.1 · Warm-up — Two-state stationarity

For $P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}$, find the stationary distribution.

Problem 15.2 · Core — Fair gambler's ruin

A fair walk starts at $k \in \{0, \dots, N\}$ and stops at 0 or N. Find the probability of hitting N first.

Problem 15.3 · Core — Expected absorption time

In the same fair walk, find the expected time to absorption.

Problem 15.4 · Advanced — Periodicity

Give a finite irreducible Markov chain with a unique stationary distribution but whose distribution from a fixed state does not converge.

Problem 15.5 · Advanced — Detailed balance

Show that if positive π satisfies $\pi_i p_{ij} = \pi_j p_{ji}$ for every i, j, then π is stationary.

Problem 15.6 · Challenge — Biased gambler's ruin

For a walk stepping right with probability $p \neq 1/2$ and left with $q = 1 - p$, stopped at $0, N$, derive the probability of hitting N before 0 from k.

Part IV

Mathematical Statistics

CHAPTER 16

Samples, Models, and Statistics

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 16.1 · Warm-up — Sample variance identity

Prove $\sum_i (X_i - \mu)^2 = \sum_i (X_i - \overline{X})^2 + n(\overline{X} - \mu)^2$.

Problem 16.2 · Core — Unbiased variance

Use the identity above to prove $S^2 = (n-1)^{-1} \sum_i (X_i - \overline{X})^2$ is unbiased for σ^2.

Problem 16.3 · Core — Empirical CDF

For fixed x, find the exact distribution, mean, and variance of $F_n(x)$.

Problem 16.4 · Advanced — Identifiability

Is the family $\{\mathcal{N}(0, \theta^2) : \theta \in \mathbb{R}\}$ identifiable? Repair the parameterization.

Problem 16.5 · Advanced — Order-statistic estimate

For $X_i \overset{\text{iid}}{\sim} \text{Unif}[0, \theta]$, compute $\mathbb{E}X_{(n)}$ and construct an unbiased estimator of θ.

Problem 16.6 · Challenge — Variance comparison

For $X_i \overset{\text{iid}}{\sim} \text{Unif}[0, \theta]$, compare the variances of the unbiased estimators $2\overline{X}$ and $(n+1)X_{(n)}/n$.

CHAPTER 17

Point Estimation

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 17.1 · Warm-up — Bernoulli MLE

For iid Bern(p) data, derive the method-of-moments estimator and the MLE of p.

Problem 17.2 · Core — Exponential rate

For iid Exp(λ) data in the rate parameterization, find the MLE and show it is consistent.

Problem 17.3 · Core — Normal MLEs

For iid $\mathcal{N}(\mu, \sigma^2)$ observations with both parameters unknown, derive their MLEs.

Problem 17.4 · Advanced — Bias–variance tradeoff

Estimator T_1 is unbiased with variance $4/n$; T_2 has bias $1/n$ and variance $2/n$. For which n does T_2 have smaller MSE?

Problem 17.5 · Advanced — Boundary likelihood

For iid Unif$[0, \theta]$ data, find the MLE and explain why solving the score equation alone fails.

Problem 17.6 · Challenge — A biased geometric MLE

A coin with unknown head probability p is tossed until the first head, and the stopping time $T = t$ is observed. Find the MLE and show it is not unbiased.

CHAPTER 18

Sufficiency and Fisher Information

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 18.1 · Warm-up — Bernoulli sufficiency

Use factorization to show $\sum_i X_i$ is sufficient for p in a Bernoulli sample.

Problem 18.2 · Core — Poisson sufficiency

For iid $\text{Pois}(\lambda)$ observations, find a sufficient statistic and the MLE.

Problem 18.3 · Core — Information

Compute Fisher information in one $\text{Exp}(\lambda)$ observation, with λ the rate.

Problem 18.4 · Advanced — Rao–Blackwell

Let $X_1, X_2 \overset{\text{iid}}{\sim} \text{Bern}(p)$ and $U = X_1$. Compute $\mathbb{E}(U \mid T)$ for $T = X_1 + X_2$ and compare its variance with U.

Problem 18.5 · Advanced — Cramér–Rao

Use Cramér–Rao to lower-bound the variance of an unbiased estimator of p from n Bernoulli observations. Does $\overline{X}$ attain it?

Problem 18.6 · Challenge — Why regularity fails

For one observation $X \sim \text{Unif}[0, \theta]$, compute the formal score away from the boundary and show its expectation is not zero. Explain the failure.

CHAPTER 19

Confidence Intervals

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 19.1 · Warm-up — Known variance

A normal sample has $n = 25$, known $\sigma = 10$, and $\overline{x} = 42$. Give a 95% confidence interval for μ using $z_{0.975} = 1.96$.

Problem 19.2 · Core — Unknown variance

Write the exact $(1 - \alpha)$ interval for the mean of a normal sample with unknown variance.

Problem 19.3 · Core — Sample-size planning

Known $\sigma = 12$. How large must n be so that a 95% interval for μ has half-width at most 2?

Problem 19.4 · Advanced — Interpretation

A reported 95% confidence interval is $[1.2, 2.8]$. Explain precisely what the 95% means and give one incorrect common interpretation.

Problem 19.5 · Advanced — Wald failure

For $n = 10$ Bernoulli trials with zero successes, compute the 95% Wald interval and explain why it is unsatisfactory.

Problem 19.6 · Challenge — A pivot for a uniform endpoint

For iid $\mathrm{Unif}[0, \theta]$ data, construct an exact $(1 - \alpha)$ confidence interval for θ using $M = X_{(n)}$.

CHAPTER 20

Hypothesis Testing

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 20.1 · Warm-up — A one-sided z test

For $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with known σ, give the level-α rejection region for $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$.

Problem 20.2 · Core — Power

For the preceding test, derive the power at true mean $\mu_1 > \mu_0$.

Problem 20.3 · Core — A p-value

A two-sided standard-normal test statistic equals 2.3. Express the p-value and state the decision at level 0.05.

Problem 20.4 · Advanced — Test–interval duality

Explain why a two-sided level-α test of $H_0 : \theta = \theta_0$ rejects exactly those θ_0 outside the corresponding $(1 - \alpha)$ confidence interval.

Problem 20.5 · Advanced — Bonferroni

You test 20 null hypotheses and want family-wise error at most 0.05. Give Bonferroni's per-test level and justify it.

Problem 20.6 · Challenge — Neyman–Pearson

One observation has distribution $\mathcal{N}(0, 1)$ under H_0 and $\mathcal{N}(1, 1)$ under H_1. Derive the most powerful level-0.05 test and its power.

CHAPTER 21

Classical Tests

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 21.1 · Warm-up — One-sample t

State the exact null distribution of $(\overline{X} - \mu_0)/(S/\sqrt{n})$ for a normal sample.

Problem 21.2 · Core — Paired design

Ten subjects are measured before and after treatment. Explain the correct test for a mean change and why an independent two-sample test is inefficient.

Problem 21.3 · Core — Goodness of fit

A die is rolled 120 times with counts $(16, 18, 20, 22, 19, 25)$. Write Pearson's statistic and its asymptotic null law.

Problem 21.4 · Advanced — Contingency table

In a 3×4 table, what are the expected counts under independence and the asymptotic degrees of freedom?

Problem 21.5 · Advanced — Welch versus pooled

When should Welch's two-sample t procedure be preferred to the pooled-variance procedure?

Problem 21.6 · Challenge — Estimated parameters and degrees of freedom

Counts in k categories are compared with probabilities $p_i(\hat{\theta})$, where an r-dimensional parameter is estimated from the same data. Explain heuristically why Pearson's asymptotic degrees of freedom become $k - 1 - r$.

CHAPTER 22

Linear Regression

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 22.1 · Warm-up — Normal equations

Differentiate $\|y - Xb\|^2$ and derive the normal equations.

Problem 22.2 · Core — Simple slope

Derive the least-squares slope and intercept in simple regression with an intercept.

Problem 22.3 · Core — Residual properties

Prove that residuals sum to zero and are uncorrelated with the fitted values when an intercept is included.

Problem 22.4 · Advanced — Adding predictors

Prove adding a predictor cannot increase training residual sum of squares.

Problem 22.5 · Advanced — Zero correlation

Give a strong nonlinear relation with zero correlation and explain what correlation measures.

Problem 22.6 · Challenge — Omitted-variable bias

The true model is $Y = \beta_0 + \beta_1 X + \gamma Z + \varepsilon$, but Z is omitted. At the population level, derive the slope from regressing Y only on X.

CHAPTER 23

Nonparametric and Asymptotic Methods

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 23.1 · Warm-up — Empirical band

Use the DKW inequality to find n sufficient for $\mathbb{P}(\sup_x |F_n - F| > 0.05) \le 0.05$.

Problem 23.2 · Core — Permutation validity

State the null assumption that makes a two-sample permutation test exact.

Problem 23.3 · Core — Bootstrap algorithm

Describe a nonparametric bootstrap estimate of the standard error of the sample median.

Problem 23.4 · Advanced — Delta method

If $\sqrt{n}(T_n - \theta) \Rightarrow \mathcal{N}(0, V)$ and $\theta > 0$, find the limit of $\sqrt{n}(\log T_n - \log \theta)$.

Problem 23.5 · Advanced — Rank invariance

Explain why ranks are unchanged by a strictly increasing transformation and why this can make rank tests robust to units.

Problem 23.6 · Challenge — Bootstrap failure at a boundary

For $X_i \sim \mathrm{Unif}[0, \theta]$, explain why the ordinary nonparametric bootstrap is problematic for the sample maximum as an estimator of θ.

Part V

Admissions Synthesis

CHAPTER 24

Admissions Problem-Solving

Learning objectives

Solve the warm-up without notes; complete both Core and Advanced problems with explicit models; attempt the Challenge as a timed admissions problem.

Problem 24.1 · Warm-up — Method recognition

A problem asks for the expected number of pairs with a property in a random configuration. State the first representation you should try and why independence is unnecessary.

Problem 24.2 · Core — Symmetry audit

What must be supplied to turn the phrase 'by symmetry, the two probabilities are equal' into a proof?

Problem 24.3 · Core — First-step template

Write the generic equations for a hitting probability h_i and expected hitting time t_i in a finite-state process with transition matrix (p_{ij}) and absorbing target states.

Problem 24.4 · Advanced — Scale check

A proposed variance formula for the average of n independent observations is $n\sigma^2$. Diagnose the error without recomputing from scratch.

Problem 24.5 · Advanced — Partial credit reduction

If you cannot evaluate a probability in closed form, list three mathematically complete intermediate forms that may still constitute substantial progress.

Problem 24.6 · Challenge — Cycles in a permutation

Find the expected number of cycles in a uniform random permutation of n elements.

Part VI

Hints and Answers

CHAPTER A

Hints and Answers

A.1 Chapter 1: Counting Before Probability

Problem 1.1

Hint. Choose positions; for the second count shift the selected positions.

Answer. $\binom{12}{4} = 495$ and $\binom{9}{4} = 126$.

Problem 1.2

Hint. Split according to the number of mathematicians.

Answer. $\binom{8}{2}\binom{6}{3} + \binom{8}{3}\binom{6}{2} = 1400$.

Problem 1.3

Hint. Remove the lower bounds first.

Answer. $\binom{17}{3} = 680$.

Problem 1.4

Hint. Count pairs (S, s) with $S \subseteq [n]$ and $s \in S$.

Answer. Choose S then s, or choose s then any subset of the remaining $n - 1$ elements.

Problem 1.5

Hint. Use inclusion–exclusion on the missing target values.

Answer. $3^n - 3 \cdot 2^n + 3$.

Problem 1.6

Hint. Track the object occupying the position of object 1 and split according to whether a two-cycle is formed.

Answer. $D_6 = 265$.

A.2 Chapter 2: Probability Spaces and Events

Problem 2.1

Hint. Use unordered hands throughout.

Answer. $\binom{4}{2}\binom{48}{3}/\binom{52}{5}$.

Problem 2.2

Hint. Assign birthdays in order.

Answer. $365 \cdot 364 \cdot 363 \cdot 362/365^4$.

Problem 2.3

Hint. Use the ratio of areas.

Answer. $F_R(r) = r^2$ and $f_R(r) = 2r$ for $0 \le r \le 1$.

Problem 2.4

Hint. Identify the increasing union.

Answer. $A_n \uparrow \Omega$ up to a null set, so continuity from below gives the claim.

Problem 2.5

Hint. Condition on the digit count.

Answer. $\mathbb{P}(7) = 1/(2 \cdot 9) = 1/18$ and $\mathbb{P}(70) = 1/(2 \cdot 90) = 1/180$.

Problem 2.6

Hint. Represent the two cut locations by a point in the unit square and remove the forbidden regions.

Answer. $1/4$.

A.3 Chapter 3: Conditional Probability and Bayes' Rule

Problem 3.1

Hint. Use the multiplication rule.

Answer. $(3/5)(2/4) = 3/10$.

Problem 3.2

Hint. Use two hypotheses and their likelihoods.

Answer. $2/3$.

Problem 3.3

Hint. False positives come from the much larger healthy group.

Answer. $0.98(0.005)/[0.98(0.005) + 0.05(0.995)] \approx 0.0897$.

Problem 3.4

Hint. The likelihoods are $(1/2)^3$ and $(3/4)^3$.

Answer. $27/43$.

Problem 3.5

Hint. Sum a geometric series over times $2, 4, 6, \ldots$.

Answer. $5/11$.

Problem 3.6

Hint. Introduce $r = \mathbb{P}(\text{say } B \mid A \text{ released})$.

Answer. The posterior is $r/(r+1)$; under the uniform rule $r = 1/2$, it is $1/3$.

A.4 Chapter 4: Independence

Problem 4.1

Hint. Subtract $\mathbb{P}(A \cap B)$ from $\mathbb{P}(B)$.

Answer. $\mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A^c)\mathbb{P}(B)$.

Problem 4.2

Hint. Use the complement and solve a logarithmic inequality.

Answer. $n \geq 21$.

Problem 4.3

Hint. Check a representative pair and then one triple atom.

Answer. Every pair is uniform on four values, but $\mathbb{P}(X = Y = Z = 0) = 1/4 \neq 1/8$.

Problem 4.4

Hint. Under the condition, only two ordered outcomes remain.

Answer. $\mathbb{P}(X = 1 \mid X + Y = 1) = 1/2$, but $\mathbb{P}(X = 1, Y = 1 \mid X + Y = 1) = 0$.

Problem 4.5

Hint. The masses increase while the ratio is at least one.

Answer. The mode is $\lfloor (n+1)p \rfloor$, with two adjacent modes if $(n+1)p$ is an integer.

Problem 4.6

Hint. Express the probabilities of all eight atoms in terms of t and impose nonnegativity.

Answer. Exactly $0 \le t \le 1/4$.

A.5 Chapter 5: Random Variables and Distribution Functions

Problem 5.1

Hint. Normalize first, then integrate piecewise.

Answer. $c = 3/4$; for $-1 \le x \le 1$, $F(x) = \frac{3}{4}(x - x^3/3 + 2/3)$.

Problem 5.2

Hint. Condition on the mixture component.

Answer. $F(x) = 0$ for $x < 0$, $F(x) = 1/3 + (2/3)x$ for $0 \le x \le 1$, and 1 above; the jump at zero is $1/3$.

Problem 5.3

Hint. For $y \ge 0$, use $-y \le X \le y$.

Answer. $F_Y(y) = F_X(y) - F_X(-y)$ for continuous X; under symmetry it is $2F_X(y) - 1$.

Problem 5.4

Hint. The inverse is $x = 1/y$ and the support reverses.

Answer. $f_Y(y) = 1/y^2$ for $1/2 \le y \le 1$.

Problem 5.5

Hint. Compute $\mathbb{P}(X \le x)$ for $x \ge 0$.

Answer. $F_X(x) = 1 - e^{-\lambda x}$.

Problem 5.6

Hint. Use one uniform variable in the first model and two in the second.

Answer. In the first model $X - Y = 0$ a.s.; in the independent model it has triangular density $1 - |z|$ on $[-1, 1]$.

A.6 Chapter 6: Expectation and Variance

Problem 6.1

Hint. Use one indicator per position.

Answer. 1.

Problem 6.2

Hint. Use one indicator per unordered pair.

Answer. $n(n-1)/4$.

Problem 6.3

Hint. Use the tail-sum formula.

Answer. 2.

Problem 6.4

Hint. Convert correlation to covariance.

Answer. $\mathrm{Cov}(X,Y) = -2$, so the variance is $16 + 9 - 4(-2) = 33$.

Problem 6.5

Hint. Randomly and independently color vertices red or blue.

Answer. Each edge crosses with probability $1/2$; average cut size is $|E|/2$.

Problem 6.6

Hint. Write $F = \sum I_i$ and compute $\mathbb{E}[I_i I_j]$ for $i \neq j$.

Answer. $\mathrm{Var}(F) = 1$.

A.7 Chapter 7: Canonical Discrete Distributions

Problem 7.1

Hint. Use a geometric law with $p = 1/6$.

Answer. Mean 6, variance 30.

Problem 7.2

Hint. Use the finite-population correction.

Answer. Mean 2; variance $10(0.2)(0.8)(90/99) = 16/11$.

Problem 7.3

Hint. Use $p_{k+1}/p_k = 4/(k+1)$.

Answer. Both 3 and 4 are modes.

Problem 7.4

Hint. The last toss is a head; choose two earlier heads.

Answer. $\binom{k-1}{2}p^3(1-p)^{k-3}$, $k \geq 3$; mean $3/p$.

Problem 7.5

Hint. Use $\lambda = np = 2$.

Answer. Approximately $1 - e^{-2}(1 + 2 + 2) = 1 - 5e^{-2}$.

Problem 7.6

Hint. Compute their joint mass by conditioning on $N = X + Y$.

Answer. $X \sim \text{Pois}(\lambda p)$, $Y \sim \text{Pois}(\lambda(1-p))$, independently.

A.8 Chapter 8: Canonical Continuous Distributions

Problem 8.1

Hint. Compute the first two moments.

Answer. $(a + b)/2$ and $(b - a)^2/12$.

Problem 8.2

Hint. Multiply survival functions.

Answer. $\min(X, Y) \sim \text{Exp}(5)$ and $\mathbb{P}(X < Y) = 2/5$.

Problem 8.3

Hint. Use a gamma law with shape three and rate λ.

Answer. $f(s) = \lambda^3 s^2 e^{-\lambda s}/2$ for $s > 0$; mean $3/\lambda$, variance $3/\lambda^2$.

Problem 8.4

Hint. The standard deviation is 2.

Answer. $\Phi(1.5) - \Phi(-1)$.

Problem 8.5

Hint. Maximum entropy is a least-structured choice relative to the stated constraints.

Answer. It justifies a conservative modeling principle, not a claim that the data-generating mechanism is normal.

Problem 8.6

Hint. Use polar coordinates or the transformation $(U,V) = (X/Y, Y)$.

Answer. $f_U(u) = 1/[\pi(1+u^2)]$.

A.9 Chapter 9: Joint Distributions and Random Vectors

Problem 9.1

Hint. Normalize by the area of the triangle.

Answer. The constant is 2; $f_X(x) = 2(1-x)$ and $f_Y(y) = 2y$ on $[0,1]$.

Problem 9.2

Hint. Compute $\mathbb{E}X, \mathbb{E}Y, \mathbb{E}XY$.

Answer. $\mathrm{Cov}(X,Y) = -1/9$; they are not independent.

Problem 9.3

Hint. Use $a^\top \Sigma a$ with $a = (2,-3)^\top$.

Answer. 85.

Problem 9.4

Hint. Use symmetry for the third moment.

Answer. $\mathbb{E}X = \mathbb{E}X^3 = 0$, so covariance is zero; Y is a deterministic function of X.

Problem 9.5

Hint. Minimize $a^2 + 4(1-a)^2$.

Answer. $a = 4/5$ and minimum variance $4/5$.

Problem 9.6

Hint. A correlation matrix must be positive semidefinite; find its eigenvalues.

Answer. $-1/2 \le r \le 1$.

A.10 Chapter 10: Transformations and Order Statistics

Problem 10.1

Hint. The event $M \le x$ requires all observations to be at most x.

Answer. $F_M(x) = x^n$, $f_M(x) = nx^{n-1}$, $\mathbb{E}M = n/(n+1)$.

Problem 10.2

Hint. Measure the overlap of $[0,1]$ and $[z-1,z]$.

Answer. $f(z) = z$ on $[0,1]$, $2-z$ on $[1,2]$, zero otherwise.

Problem 10.3

Hint. Use survival functions and one integral for the winning clock.

Answer. Minimum rate $\sum_i \lambda_i$; winner probability $\lambda_j / \sum_i \lambda_i$.

Problem 10.4

Hint. The median is the third order statistic.

Answer. $30F(x)^2[1-F(x)]^2 f(x)$.

Problem 10.5

Hint. Condition on N and use total variance.

Answer. Mean 8, variance $4 \cdot 5 + 3 \cdot 2^2 = 32$.

Problem 10.6

Hint. Use $x = uv, y = u(1-v)$ and compute the Jacobian.

Answer. $U \sim \text{Gamma}(2,1)$ and $V \sim \text{Unif}[0,1]$, independently.

A.11 Chapter 11: Conditional Expectation

Problem 11.1

Hint. Condition on the die value.

Answer. $\mathbb{E}(N/2) = 7/4$.

Problem 11.2

Hint. Use total expectation and total variance.

Answer. Mean $3/2$; variance $(1+4)/2 + \text{Var}(0,3) = 5/2 + 9/4 = 19/4$.

Problem 11.3

Hint. Conditional Poisson mean and variance both equal Λ.

Answer. Mean 2, variance $2 + 3 = 5$.

Problem 11.4

Hint. Condition the product on Y and pull out $g(Y)$.

Answer. The conditional expectation is $g(Y)[\mathbb{E}(X \mid Y) - m(Y)] = 0$.

Problem 11.5

Hint. The maximum among the first k positions is equally likely to occupy any of them.

Answer. $H_n = \sum_{k=1}^{n} 1/k$.

Problem 11.6

Hint. Use states corresponding to matched prefixes: none, H, and HT.

Answer. 10.

A.12 Chapter 12: Probability Inequalities

Problem 12.1

Hint. Apply Markov directly.

Answer. At most $1/5$.

Problem 12.2

Hint. $\mathrm{Var}(\overline{X}) = 9/n$.

Answer. $n \geq 3600$.

Problem 12.3

Hint. Use Jensen on $x \mapsto 1/x$ or Cauchy–Schwarz.

Answer. Equality holds exactly when X is constant almost surely.

Problem 12.4

Hint. Take a union over pairs.

Answer. At most $\binom{n}{2}/365$.

Problem 12.5

Hint. Apply Markov to e^{tS} and factor the MGF.

Answer. The displayed inequality follows for every $t > 0$; minimize over t.

Problem 12.6

Hint. Use a random vertex order and select local minima.

Answer. Select vertices preceding all neighbors; expectation is $\sum_v 1/(d(v)+1) \geq n/(d+1)$.

A.13 Chapter 13: Generating and Characteristic Functions

Problem 13.1

Hint. Represent X as a sum of Bernoulli variables.

Answer. $G_X(s) = (1-p+ps)^n$ and $G'_X(1) = np$.

Problem 13.2

Hint. Multiply $e^{\lambda(s-1)}$ and $e^{\mu(s-1)}$.

Answer. The product is $e^{(\lambda+\mu)(s-1)}$.

Problem 13.3

Hint. Condition on N.

Answer. $G_S(s) = \exp\{\lambda(G(s)-1)\}$.

Problem 13.4

Hint. Expand as a geometric series and differentiate.

Answer. $\mathbb{P}(X=k) = (1-q)q^k$, $k \geq 0$; mean $q/(1-q)$, variance $q/(1-q)^2$.

Problem 13.5

Hint. Take the limit of $(1+\lambda(s-1)/n)^n$.

Answer. The limit is $e^{\lambda(s-1)}$.

Problem 13.6

Hint. Let q_n be the probability of extinction by generation n and condition on the first generation.

Answer. $q_{n+1} = G(q_n)$ with $q_0 = 0$, so $q_n \uparrow q = G(q)$; monotonicity shows q is the smallest fixed point.

A.14 Chapter 14: Convergence and Limit Theorems

Problem 14.1

Hint. Apply Chebyshev to the sample mean.

Answer. At most $16/n$; take $n \geq 1600$.

Problem 14.2

Hint. For fixed ε, evaluate the exceptional probability.

Answer. $X_n \to 0$ in probability, but $\mathbb{E}X_n = 1$.

Problem 14.3

Hint. Standard deviation is 5; use endpoints 44.5 and 55.5.

Answer. Approximately $\Phi(1.1) - \Phi(-1.1)$.

Problem 14.4

Hint. Use continuity of $x \mapsto 1/x$ at θ or an explicit neighborhood bound.

Answer. It follows from the continuous mapping theorem.

Problem 14.5

Hint. Apply the CLT without dividing by the standard deviation.

Answer. $\mathcal{N}(0, 4)$.

Problem 14.6

Hint. Use iid copies of a Bernoulli variable and compare each with one fixed copy.

Answer. Let $X, X_1, X_2, \ldots$ be iid Bern$(1/2)$. Then $X_n \Rightarrow X$, but $\mathbb{P}(|X_n - X| > 1/2) = 1/2$.

A.15 Chapter 15: Random Walks and Markov Chains

Problem 15.1

Hint. Solve $\pi_1(0.2) = \pi_2(0.3)$ and normalize.

Answer. $\pi = (0.6, 0.4)$.

Problem 15.2

Hint. Solve the harmonic recurrence with two boundary values.

Answer. k/N.

Problem 15.3

Hint. Solve $t_k = 1 + (t_{k-1} + t_{k+1})/2$ with zero boundaries.

Answer. $k(N - k)$.

Problem 15.4

Hint. Use deterministic alternation between two states.

Answer. $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; stationary $(1/2, 1/2)$, period two.

Problem 15.5

Hint. Sum detailed-balance identities over the incoming state.

Answer. $\sum_i \pi_i p_{ij} = \sum_i \pi_j p_{ji} = \pi_j$.

Problem 15.6

Hint. Solve $h_k = p h_{k+1} + q h_{k-1}$ using a geometric trial solution.

Answer. $h_k = [1 - (q/p)^k]/[1 - (q/p)^N]$.

A.16 Chapter 16: Samples, Models, and Statistics

Problem 16.1

Hint. Expand after adding and subtracting $\overline{X}$.

Answer. The cross term vanishes because $\sum_i (X_i - \overline{X}) = 0$.

Problem 16.2

Hint. Take expectations and use $\mathrm{Var}(\overline{X}) = \sigma^2/n$.

Answer. The expected numerator is $(n-1)\sigma^2$.

Problem 16.3

Hint. It is an average of Bernoulli indicators.

Answer. $nF_n(x) \sim \mathrm{Bin}(n, F(x))$; mean $F(x)$, variance $F(x)(1 - F(x))/n$.

Problem 16.4

Hint. Compare θ and $-\theta$.

Answer. No; restrict to $\theta \geq 0$ or parameterize by variance $v \geq 0$.

Problem 16.5

Hint. Scale the maximum of n uniforms on $[0, 1]$.

Answer. $\mathbb{E}X_{(n)} = n\theta/(n+1)$; use $(n+1)X_{(n)}/n$.

Problem 16.6

Hint. Use $\mathrm{Var}(X) = \theta^2/12$ and the beta moments of the maximum.

Answer. Variances are $\theta^2/(3n)$ and $\theta^2/[n(n+2)]$; the maximum-based estimator is better for $n > 1$.

A.17 Chapter 17: Point Estimation

Problem 17.1

Hint. The first population moment is p; maximize the binomial likelihood.

Answer. Both are $\hat{p} = \overline{X}$.

Problem 17.2

Hint. Differentiate $n \log \lambda - \lambda \sum X_i$; then use the LLN and continuous mapping.

Answer. $\hat{\lambda} = 1/\overline{X} \to \lambda$ in probability.

Problem 17.3

Hint. Optimize first in μ, then in σ^2.

Answer. $\hat{\mu} = \overline{X}$ and $\hat{\sigma}^2 = n^{-1} \sum_i (X_i - \overline{X})^2$.

Problem 17.4

Hint. MSE is variance plus squared bias.

Answer. For every positive integer n: $2/n + 1/n^2 < 4/n$.

Problem 17.5

Hint. Include the support indicator in the likelihood.

Answer. $\hat{\theta} = X_{(n)}$; the maximum occurs at the boundary of the feasible parameter set.

Problem 17.6

Hint. Maximize $p(1-p)^{t-1}$. To check bias, evaluate $\mathbb{E}(1/T)$ or compare its first terms with p.

Answer. $\hat{p} = 1/T$; for $0 < p < 1$, $\mathbb{E}(1/T) = -p \log p/(1-p) \neq p$.

A.18 Chapter 18: Sufficiency and Fisher Information

Problem 18.1

Hint. Write the joint mass as $p^{\sum x_i}(1-p)^{n-\sum x_i}$.

Answer. All parameter dependence is through $\sum_i x_i$.

Problem 18.2

Hint. Factor the joint likelihood.

Answer. $T = \sum_i X_i$ is sufficient; $\hat{\lambda} = \overline{X}$.

Problem 18.3

Hint. The score is $1/\lambda - X$.

Answer. $I(\lambda) = 1/\lambda^2$.

Problem 18.4

Hint. Given T, symmetry makes each coordinate have conditional mean $T/2$.

Answer. $U^* = T/2 = \overline{X}$; its variance is $p(1-p)/2$, half that of U.

Problem 18.5

Hint. Information per observation is $1/[p(1-p)]$.

Answer. Bound $p(1-p)/n$; yes, $\overline{X}$ attains it.

Problem 18.6

Hint. Differentiate $-\log\theta$ while noting the moving support.

Answer. The interior score is $-1/\theta$, whose expectation is not zero; differentiation under the integral ignores a boundary contribution.

A.19 Chapter 19: Confidence Intervals

Problem 19.1

Hint. Standard error is $10/5$.

Answer. 42 ± 3.92, i.e. $[38.08, 45.92]$.

Problem 19.2

Hint. Use the Student pivot.

Answer. $\overline{X} \pm t_{n-1,1-\alpha/2} S/\sqrt{n}$.

Problem 19.3

Hint. Solve $1.96\sigma/\sqrt{n} \le 2$.

Answer. $n \ge 139$.

Problem 19.4

Hint. Refer to repetition of the full sampling procedure.

Answer. The procedure covers the fixed parameter in 95% of repeated samples; it is not generally true that the fixed parameter has posterior probability 0.95 of lying in this observed interval.

Problem 19.5

Hint. Substitute $\hat{p} = 0$ in the usual standard error.

Answer. It degenerates to $[0, 0]$, despite substantial uncertainty; score or exact intervals are preferable.

Problem 19.6

Hint. The ratio M/θ has CDF u^n on $[0, 1]$.

Answer. $[M, M/\alpha^{1/n}]$ has exact coverage $1 - \alpha$.

A.20 Chapter 20: Hypothesis Testing

Problem 20.1

Hint. Standardize the sample mean under the null.

Answer. Reject when $(\overline{X} - \mu_0)/(\sigma/\sqrt{n}) > z_{1-\alpha}$.

Problem 20.2

Hint. Express the rejection threshold in units centered at μ_1.

Answer. $1 - \Phi(z_{1-\alpha} - (\mu_1 - \mu_0)\sqrt{n}/\sigma)$.

Problem 20.3

Hint. Double the upper-tail probability.

Answer. $p = 2[1 - \Phi(2.3)] \approx 0.0214$; reject.

Problem 20.4

Hint. The interval is the set of parameter values whose tests accept the observed data.

Answer. It follows by inversion of the acceptance regions.

Problem 20.5

Hint. Use a union bound over false rejections.

Answer. Use 0.0025 per test; the total is at most $20(0.0025) = 0.05$.

Problem 20.6

Hint. The likelihood ratio is increasing in X.

Answer. Reject for $X > z_{0.95}$; power $1 - \Phi(z_{0.95} - 1)$.

A.21 Chapter 21: Classical Tests

Problem 21.1

Hint. Count the estimated variance degrees of freedom.

Answer. Student t_{n-1}.

Problem 21.2

Hint. Reduce the data to within-subject differences.

Answer. Use a one-sample t test on $D_i = \mathrm{after}_i - \mathrm{before}_i$; pairing uses the within-subject covariance.

Problem 21.3

Hint. Every expected count is 20.

Answer. $X^2 = \sum_i (O_i - 20)^2/20$ and is approximately χ^2_5 under fairness.

Problem 21.4

Hint. Use row total times column total divided by grand total.

Answer. $E_{ij} = R_i C_j / n$ and degrees of freedom $(3-1)(4-1) = 6$.

Problem 21.5

Hint. Focus on the equal-variance assumption.

Answer. Welch is preferred when variances may differ; it remains reliable under equality with only a small power cost.

Problem 21.6

Hint. Start from k residuals, remove the total-count constraint and the fitted parameter directions.

Answer. One degree is lost because residuals sum to zero and r more because fitting forces orthogonality to r score directions.

A.22 Chapter 22: Linear Regression

Problem 22.1

Hint. Expand the quadratic or use matrix calculus.

Answer. $X^\top X \hat{b} = X^\top y$.

Problem 22.2

Hint. Center both variables first.

Answer. $\hat{\beta}_1 = \sum(x_i - \bar{x})(y_i - \bar{y}) / \sum(x_i - \bar{x})^2$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.

Problem 22.3

Hint. Residuals are orthogonal to every column of X and hence to the column space.

Answer. $\mathbf{1}^\top e = 0$ and $\hat{y}^\top e = 0$.

Problem 22.4

Hint. The old fitted vectors remain available in the enlarged model.

Answer. The minimum over a superset cannot exceed the old minimum.

Problem 22.5

Hint. Use $Y = X^2$ with symmetric X.

Answer. Then $\operatorname{Cov}(X, Y) = 0$ but Y is determined by X; correlation measures linear association.

Problem 22.6

Hint. Use $\text{Slope} = \operatorname{Cov}(X, Y) / \operatorname{Var}(X)$.

Answer. $\beta_1 + \gamma \operatorname{Cov}(X, Z) / \operatorname{Var}(X)$.

A.23 Chapter 23: Nonparametric and Asymptotic Methods

Problem 23.1

Hint. Solve $2e^{-2n(0.05)^2} \leq 0.05$.

Answer. $n \geq \lceil \log(40)/0.005 \rceil = 738$.

Problem 23.2

Hint. Conditional on pooled observations, labels must be exchangeable.

Answer. The joint distribution must be invariant under the allowed label permutations.

Problem 23.3

Hint. Resample the observed data with replacement many times.

Answer. Compute the median in each size-n bootstrap sample and take the empirical standard deviation of those medians.

Problem 23.4

Hint. Derivative of $\log x$ at θ is $1/\theta$.

Answer. $\mathcal{N}(0, V/\theta^2)$.

Problem 23.5

Hint. Increasing maps preserve every pairwise ordering.

Answer. All ranks, and hence rank-test statistics, are identical before and after the transformation.

Problem 23.6

Hint. Every bootstrap sample is drawn from observed points and can never exceed the observed maximum.

Answer. The bootstrap distribution has an atom at the observed maximum and cannot reproduce new observations in $(X_{(n)}, \theta]$; the estimator's boundary-scale behavior is misrepresented.

A.24 Chapter 24: Admissions Problem-Solving

Problem 24.1

Hint. Use one indicator per pair.

Answer. Write $N = \sum_{i<j} I_{ij}$ and use $\mathbb{E}N = \sum_{i<j} \mathbb{P}(I_{ij} = 1)$; linearity needs no independence.

Problem 24.2

Hint. Identify a probability-preserving map.

Answer. Give a bijection or measure-preserving transformation of outcomes mapping one event onto the other.

Problem 24.3

Hint. Condition on the first transition.

Answer. $h_i = \sum_j p_{ij} h_j$ and $t_i = 1 + \sum_j p_{ij} t_j$, with boundary values $h = 0$ or 1 and $t = 0$ at absorbing states.

Problem 24.4

Hint. Check how variance scales under multiplication by $1/n$.

Answer. The sum has variance $n\sigma^2$, and averaging multiplies variance by $1/n^2$, giving σ^2/n.

Problem 24.5

Hint. Think exact sum, recurrence, or bound.

Answer. A justified finite/infinite sum, a recurrence with boundary conditions, or a valid upper/lower bound or asymptotic approximation with checked conditions.

Problem 24.6

Hint. Insert elements one at a time and indicate whether a new cycle begins.

Answer. $H_n = 1 + 1/2 + \cdots + 1/n$.

Part VII

Selected Complete Solutions

CHAPTER B

Selected Complete Solutions

B.1 Problem 1.6: Derangements recurrence

Let D_n be the number of derangements of n objects. Prove $D_n = (n-1)(D_{n-1} + D_{n-2})$ and use it to compute D_6.

Solution. Let $j = \pi(1)$, so $j \neq 1$, giving $n-1$ choices. If $\pi(j) = 1$, the pair $(1, j)$ is a two-cycle and the remaining objects form a derangement, giving D_{n-2}. Otherwise, remove object 1 and redirect the arrow into 1 toward j; this is a bijection with derangements of $n-1$ objects. Thus each j contributes $D_{n-2} + D_{n-1}$. Starting with $D_1 = 0, D_2 = 1$, the recurrence gives $D_3 = 2, D_4 = 9, D_5 = 44, D_6 = 265$.

B.2 Problem 2.6: Broken stick

Two points are chosen independently and uniformly on a unit segment. The three pieces form a triangle exactly when no piece exceeds $1/2$. Find the probability.

Solution. Let the cut locations be (X, Y), uniform on the unit square, and assume $X < Y$ by symmetry. The piece lengths are $X, Y - X, 1 - Y$. The conditions $X < 1/2$, $Y - X < 1/2$, and $1 - Y < 1/2$ define a central triangle inside the half-square $0 < X < Y < 1$. Its area is $1/8$. The reflected region for $Y < X$ has the same area, so the total favorable area is $1/4$.

B.3 Problem 3.6: Information protocol

One of prisoners A, B, C will be released uniformly. A guard truthfully names one of B, C who will not be released and says B. Show that the posterior probability for A cannot be determined unless the guard's rule is specified; compute it if the guard chooses uniformly when both names are available.

Solution. If A is released, both B and C are valid names, and let the probability of naming B be r. If B is released, the guard must name C, so the likelihood of hearing B is zero. If C is released, the guard must name B, so the likelihood is one. Bayes' rule gives $\mathbb{P}(A \mid \text{said } B) = (r/3)/(r/3 + 1/3) = r/(r+1)$. The answer therefore depends on the protocol. For $r = 1/2$, it equals $1/3$.

B.4 Problem 4.6: Possible triple intersections

Events A, B, C are pairwise independent and each has probability $1/2$. Determine all possible values of $t = \mathbb{P}(A \cap B \cap C)$.

Solution. Pairwise independence fixes every two-way intersection at $1/4$. Starting with the atom ABC of mass t, each atom with exactly two named events has mass $1/4 - t$. The atoms with exactly one event have mass t, and the atom with none has mass $1/4 - t$. These eight masses sum to one and are nonnegative exactly for $0 \leq t \leq 1/4$. Conversely, assigning these masses constructs a valid model for every such t.

B.5 Problem 5.6: Same distribution, different dependence

Construct random variables X, Y with the same Unif$[0,1]$ marginal distribution in two models: one where $X = Y$ almost surely and one where X, Y are independent. Compare the distribution of $X - Y$.

Solution. For the dependent model take $U \sim$ Unif$[0,1]$ and set $X = Y = U$. For the independent model take independent $U, V \sim$ Unif$[0,1]$ and set $X = U, Y = V$. Both coordinates have the same marginal CDF in both models. In the first, the difference is identically zero. In the second, convolution with $-V$ gives $f_{U-V}(z) = 1 - |z|$ for $|z| \le 1$. Marginals do not determine a joint law.

B.6 Problem 6.6: Variance of fixed points

Let F be the number of fixed points in a uniform random permutation of $n \ge 2$ elements. Compute $\mathrm{Var}(F)$.

Solution. We have $\mathbb{E}I_i = 1/n$ and $\mathrm{Var}(I_i) = 1/n - 1/n^2$. For $i \ne j$, $\mathbb{E}(I_iI_j) = \mathbb{P}(\pi(i) = i, \pi(j) = j) = (n-2)!/n! = 1/[n(n-1)]$. Hence $\mathrm{Cov}(I_i, I_j) = 1/[n(n-1)] - 1/n^2 = 1/[n^2(n-1)]$. Therefore $\mathrm{Var}(F) = n(1/n - 1/n^2) + 2\binom{n}{2}/[n^2(n-1)] = (1 - 1/n) + 1/n = 1$.

B.7 Problem 7.6: Poisson splitting

Let $N \sim$ Pois(λ). Independently label each of the N items type A with probability p and type B otherwise. Prove the two type counts are independent Poisson variables.

Solution. For $x, y \ge 0$, the event $(X, Y) = (x, y)$ requires $N = x + y$ and a binomial split. Thus its probability is $e^{-\lambda}\lambda^{x+y}/(x+y)! \cdot \binom{x+y}{x}p^x(1-p)^y$. Rearranging gives $[e^{-\lambda p}(\lambda p)^x/x!][e^{-\lambda(1-p)}(\lambda(1-p))^y/y!]$, a product of the two marginal Poisson masses. Hence the counts are independent.

B.8 Problem 8.6: Ratio of normals

Let X, Y be independent standard normal variables. Show that X/Y has the standard Cauchy density.

Solution. Use $x = uv, y = v$, whose Jacobian magnitude is $|v|$. Then $f_{U,V}(u, v) = (2\pi)^{-1}\exp[-(1+u^2)v^2/2]|v|$. Integrating over $v \in \mathbb{R}$ gives $f_U(u) = (2\pi)^{-1} \cdot 2/(1+u^2) = 1/[\pi(1+u^2)]$. The heavy tail also explains why the Cauchy mean does not exist.

B.9 Problem 9.6: A covariance matrix criterion

For which real r can $\begin{pmatrix} 1 & r & r \\ r & 1 & r \\ r & r & 1 \end{pmatrix}$ be a correlation matrix?

Solution. The vector $(1, 1, 1)$ is an eigenvector with eigenvalue $1 + 2r$. Every vector orthogonal to it is an eigenvector with eigenvalue $1 - r$, of multiplicity two. Positive semidefiniteness is therefore equivalent to $1 + 2r \ge 0$ and $1 - r \ge 0$, giving $-1/2 \le r \le 1$. These values are attainable, for example by a centered Gaussian vector with this covariance matrix.

B.10 Problem 10.6: Independence after a change of variables

Independent $X, Y \sim \text{Exp}(1)$ are transformed to $U = X + Y$ and $V = X/(X+Y)$. Prove U, V are independent and identify their laws.

Solution. The inverse transformation has support $u > 0, 0 < v < 1$ and Jacobian u. Hence $f_{U,V}(u,v) = e^{-uv}e^{-u(1-v)}u = ue^{-u}$. This is the product of the gamma density ue^{-u} on $u > 0$ and the uniform density 1 on $0 < v < 1$. Factorization on a product support proves independence.

B.11 Problem 11.6: Waiting for HTH

A fair coin is tossed until the pattern HTH first appears. Find the expected number of tosses.

Solution. Let e_0, e_1, e_2 be expected additional tosses after matching prefixes of lengths $0, 1, 2$. First-step analysis gives $e_0 = 1 + \frac{1}{2}e_1 + \frac{1}{2}e_0$, $e_1 = 1 + \frac{1}{2}e_1 + \frac{1}{2}e_2$ (HH leaves suffix H), and $e_2 = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}e_0$ (HTT has no useful suffix). Thus $e_0 = 2 + e_1$, $e_1 = 2 + e_2$, and $e_2 = 1 + e_0/2$. Solving yields $e_0 = 10$.

B.12 Problem 12.6: A large independent set

Prove every graph with n vertices and average degree d contains an independent set of size at least $n/(d+1)$.

Solution. Choose a uniform random ordering of the vertices and include v if it occurs before all its neighbors. Adjacent vertices cannot both be selected, so the selected set is independent. Among v and its $d(v)$ neighbors, each is equally likely to be first, hence $\mathbb{P}(v \text{ selected}) = 1/(d(v)+1)$. The expected size is $\sum_v 1/(d(v)+1)$. Since $x \mapsto 1/(x+1)$ is convex, Jensen gives at least $n/(d+1)$. Some ordering attains at least the expectation.

B.13 Problem 13.6: Branching extinction

In a Galton–Watson process each individual has offspring PGF G. Prove the extinction probability q is the smallest fixed point of G in $[0,1]$.

Solution. Let q_n be the probability that the population is extinct by generation n, starting with one ancestor. We have $q_0 = 0$. If the ancestor has k children, extinction by generation $n+1$ requires all k descendant processes to be extinct by generation n, with probability q_n^k. Hence $q_{n+1} = G(q_n)$. The events increase, so $q_n \uparrow q$, and continuity gives $q = G(q)$. If $r \in [0,1]$ is any fixed point, induction from $q_0 = 0 \le r$ gives $q_n \le r$, hence $q \le r$.

B.14 Problem 14.6: Distribution but not probability

Give a sequence X_n that converges in distribution but not in probability to a random variable with the same nondegenerate law.

Solution. All X_n have exactly the same distribution as X, so convergence in distribution is immediate. However, because X_n and X are independent fair bits, they differ with probability $1/2$ for every n. Hence the probability of a discrepancy larger than $1/2$ does not tend to zero, and there is no convergence in probability.

B.15 Problem 15.6: Biased gambler's ruin

For a walk stepping right with probability $p \neq 1/2$ and left with $q = 1-p$, stopped at $0, N$, derive the probability of hitting N before 0 from k.

Solution. The recurrence rearranges to $p(h_{k+1} - h_k) = q(h_k - h_{k-1})$. Thus successive differences form a geometric sequence with ratio q/p, so $h_k = A + B(q/p)^k$. Boundary condition $h_0 = 0$ gives $A = -B$; $h_N = 1$ gives $B[(q/p)^N - 1] = 1$. Substitution yields the stated formula. Its limit as $p \to 1/2$ is k/N.

B.16 Problem 16.6: Variance comparison

For $X_i \overset{\text{iid}}{\sim} \text{Unif}[0, \theta]$, compare the variances of the unbiased estimators $2\overline{X}$ and $(n+1)X_{(n)}/n$.

Solution. Since $\text{Var}(\overline{X}) = \theta^2/(12n)$, $\text{Var}(2\overline{X}) = \theta^2/(3n)$. For $M = X_{(n)}$, $M/\theta \sim \text{Beta}(n, 1)$, hence $\text{Var}(M) = \theta^2 n/[(n+1)^2(n+2)]$. Multiplying by $((n+1)/n)^2$ gives $\theta^2/[n(n+2)]$. The latter is smaller than $\theta^2/(3n)$ exactly when $n > 1$.

B.17 Problem 17.6: A biased geometric MLE

A coin with unknown head probability p is tossed until the first head, and the stopping time $T = t$ is observed. Find the MLE and show it is not unbiased.

Solution. The log-likelihood is $\log p + (t-1)\log(1-p)$, whose derivative vanishes at $p = 1/t$ and whose second derivative is negative. Next, $\mathbb{E}(1/T) = \sum_{t \geq 1} p(1-p)^{t-1}/t$. Using $\sum_{t \geq 1} q^t/t = -\log(1-q)$ with $q = 1-p$, this equals $-p\log p/(1-p)$, which is not p except at limiting boundary cases. MLE invariance and consistency do not imply unbiasedness.

B.18 Problem 18.6: Why regularity fails

For one observation $X \sim \text{Unif}[0, \theta]$, compute the formal score away from the boundary and show its expectation is not zero. Explain the failure.

Solution. For $0 < x < \theta$, $\partial_\theta \log f_\theta(x) = -1/\theta$, so its expectation is $-1/\theta$. The familiar identity $\mathbb{E}U_\theta = 0$ would require interchanging differentiation and integration over a fixed support. Here $\int_0^\theta \theta^{-1} dx = 1$ has a moving upper limit; differentiating only the integrand misses the boundary term. Consequently, the standard Cramér–Rao theorem cannot be applied mechanically.

B.19 Problem 19.6: A pivot for a uniform endpoint

For iid $\text{Unif}[0, \theta]$ data, construct an exact $(1-\alpha)$ confidence interval for θ using $M = X_{(n)}$.

Solution. Since $\mathbb{P}(M/\theta \leq u) = u^n$, we have $\mathbb{P}(M/\theta \geq \alpha^{1/n}) = 1 - \alpha$. The event $M/\theta \geq \alpha^{1/n}$ is equivalent to $\theta \leq M/\alpha^{1/n}$, which is an upper bound, not a lower bound. Thus an exact upper confidence bound is $U = M/\alpha^{1/n}$. A nontrivial finite lower bound cannot improve on M, because the support itself forces $\theta \geq M$ with probability one. The exact interval $[M, M/\alpha^{1/n}]$ has coverage $1 - \alpha$.

B.20 Problem 20.6: Neyman–Pearson

One observation has distribution $\mathcal{N}(0,1)$ under H_0 and $\mathcal{N}(1,1)$ under H_1. Derive the most powerful level-0.05 test and its power.

Solution. The likelihood ratio is $\exp(x-1/2)$, hence increases with x. Neyman–Pearson therefore chooses an upper-tail region. Under H_0, selecting $c = z_{0.95}$ makes $\mathbb{P}_0(X > c) = 0.05$. Under H_1, $X-1$ is standard normal, so the power is $\mathbb{P}_1(X > c) = 1-\Phi(c-1) = 1-\Phi(z_{0.95}-1)$.

B.21 Problem 21.6: Estimated parameters and degrees of freedom

Counts in k categories are compared with probabilities $p_i(\hat{\theta})$, where an r-dimensional parameter is estimated from the same data. Explain heuristically why Pearson's asymptotic degrees of freedom become $k-1-r$.

Solution. The vector of standardized count deviations initially lies in a k-dimensional space. Since the counts sum to n, deviations satisfy one linear constraint, leaving dimension $k-1$. Estimating r regular parameters uses the data to fit r independent directions in this residual space. The remaining unexplained component has dimension $k-1-r$, and its squared asymptotic Gaussian length gives a χ^2_{k-1-r} limit.

B.22 Problem 22.6: Omitted-variable bias

The true model is $Y = \beta_0 + \beta_1 X + \gamma Z + \varepsilon$, but Z is omitted. At the population level, derive the slope from regressing Y only on X.

Solution. Taking covariance with X gives $\operatorname{Cov}(X,Y) = \beta_1 \operatorname{Var}(X) + \gamma \operatorname{Cov}(X,Z) + \operatorname{Cov}(X,\varepsilon)$. Under the true exogeneity condition $\operatorname{Cov}(X,\varepsilon) = 0$, division by $\operatorname{Var}(X)$ yields the simple-regression slope $\beta_1 + \gamma \operatorname{Cov}(X,Z)/\operatorname{Var}(X)$. Bias appears when the omitted variable affects Y and is correlated with X.

B.23 Problem 23.6: Bootstrap failure at a boundary

For $X_i \sim \mathrm{Unif}[0,\theta]$, explain why the ordinary nonparametric bootstrap is problematic for the sample maximum as an estimator of θ.

Solution. Conditional on the observed sample, every bootstrap observation is at most $M = X_{(n)}$, so every bootstrap maximum M^* is at most M and equals M with substantial probability. In the real experiment, a new sample maximum can lie above the current M up to the unknown endpoint θ. The empirical distribution has the wrong endpoint and therefore cannot mimic the $n(\theta - M)$ limit. Parametric bootstrap or endpoint-specific methods are needed.

B.24 Problem 24.6: Cycles in a permutation

Find the expected number of cycles in a uniform random permutation of n elements.

Solution. Construct a uniform permutation sequentially in cycle notation. When inserting element k, there are k equally likely insertion locations: after each of the previous $k-1$ elements, or as a new singleton cycle. Exactly one location starts a new cycle. Let I_k indicate that event; then

$\mathbb{P}(I_k = 1) = 1/k$. Every cycle has exactly one element that started it, so the cycle count is $C_n = \sum_{k=1}^{n} I_k$. Linearity yields $\mathbb{E}C_n = \sum_{k=1}^{n} 1/k = H_n$.

Three Mock Examinations

Mock A — foundations, 120 minutes

Problems 1.3, 2.4, 3.3, 4.5, 6.2, and 7.4.

Mock B — probability synthesis, 180 minutes

Problems 5.5, 8.4, 9.5, 10.6, 11.4, and 12.5.

Mock C — statistics and data reasoning, 180 minutes

Problems 16.4, 17.5, 18.4, 19.5, 20.6, and 22.5.

For each mock, submit only complete arguments. After grading, classify every loss as a modeling error, theorem-selection error, algebraic error, or exposition gap.